

The Filtered MOND Action: a Central Tidal Identity, Comparable-Mass Forces, a First Covariant Clock Action, and the Operator That Screens the Solar System

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Abstract

We collect, verify and extend the results landed in the closure programme of the de Sitter–MOND repository between 4 and 5 September 2026 for the *filtered* two-potential MOND action (a spatially filtered specialisation of QUMOND with the exact inverse partner of $\mu(y) = 1 - e^{-y}$ and one coherence length ξ). (i) The action’s linear response about a constant external field gives a central tidal identity, $Q_2/D = 2y/\{5[3\lambda(y) - y]\}$, $\lambda = 1 + (y - 1)e^{-y}$, independent of source mass and filter amplitude; we reproduce it with independent nonlinear machinery to 1.4% (canonical footing) and 0.6% (alternate), with the mass and ξ independence holding to 5×10^{-5} . (ii) Varying both source positions gives the comparable-mass force; for an isolated compact filtered pair its leading harmonic coefficient $h = \frac{2}{9}\sqrt{GMa_0}/\xi^2$ is reproduced to 2–5%, converging as ξ/r_M grows, and in the external-field regime the identity becomes a three-frequency relation $2y(\omega_r^2 - \omega_\phi^2) = 3[5\lambda - y](\omega_\phi^2 - \omega_z^2)$. (iii) The exact two-body force from the stationary action reproduces Milgrom’s deep-MOND two-body law to 1%, verifies that a bounded body’s phantom exerts no net force on it in a uniform external field, and shows that the total-mass one-body reduction used by external-field wide-binary pipelines errs by under 2.5% except for an 8–10% spike at $d = \sqrt{GM_{\text{tot}}/g_e}$ (aligned geometry only). (iv) Solar-System gates: the double filter survives for $\xi \geq 0.02\text{--}0.03$ pc. (v) A first explicit covariant clock/heat action (C-H) reduces at leading weak static order to the filtered functional with both filters; its physical mode count and causal feasibility remain open, and an intrinsic-elliptic Deffayet–Woodard variant is excluded exactly because it admits no expanding FLRW background. (vi) The local term $\xi^2|\nabla_\perp V|^2$ added to the scalar’s gradient invariant Y inside the MOND function, $V_\mu = q_\mu{}^\nu\partial_\nu\phi$, reproduces the coherent stiffening $J_Y \rightarrow J_Y(1 + \xi^2k^2)$ exactly in α_1 in both an aether host and a hypersurface-orthogonal clock host, suppresses the scalar’s α_2 drag by 3×10^{-5} at $(\xi k)^2 = 10^4$, adds no degree of freedom, and leaves the static PPN ladder under its bounds at the Cassini floor. Nothing here is a completed relativistic theory, a derivation of ξ or of a_0 , or an empirical detection; the wide-binary pre-registration is untouched.

1 The filtered action

With u the Newtonian potential, Φ the potential felt by matter, $S = e^{\xi^2\Delta/2}$ a Gaussian filter of standard deviation ξ (or the Helmholtz filter $(1 - \xi^2\Delta)^{-1}$), and $q'(s^2) = \nu(s) - 1$ where ν is the exact inverse partner of $\mu(y) = 1 - e^{-y}$ ($s = y\mu(y)$, $\nu = y/s$),

$$I = \int dt d^3x \left\{ L_m - \rho\Phi - \frac{2\nabla\Phi \cdot \nabla u - |\nabla u|^2 - a_0^2 q(|\nabla Su|^2/a_0^2)}{8\pi G} \right\}. \quad (1)$$

Variation gives $\Delta u = 4\pi G\rho$ and $\Delta\Phi = 4\pi G\rho + S\nabla\cdot[(\nu-1)\nabla Su]$: the outer (adjoint) filter is forced by the variation, not assumed. The on-shell energy is $E = -(8\pi G)^{-1}\int |\nabla u|^2 - (a_0^2/8\pi G)\int Q_t(|\nabla Su|/a_0)$ with the exact primitive $Q_t(s) = y^2 + 2 - (2y^2 + 2y + 2)e^{-y} - s^2$, $y = s\nu(s)$. Both footings $a_0 = 9.3619 \times 10^{-11}$ and $1.1279 \times 10^{-10} \text{ m s}^{-2}$ are carried throughout; $\kappa = \frac{1}{2}$ in $a_0 = \kappa c\sqrt{G\rho_\Lambda}$ is fitted, not derived.

2 The central tidal identity

Linearising about a constant filtered Newtonian background $s_e a_0$ and adding a spherical source, the additional central Hessian $H_{ij} = \partial_i \partial_j \Phi_{\text{ph}}(0)$ has trace $D = 4\pi G\rho_{\text{ph}}(0)$ and axial quadrupole $Q_2 = H_\perp - H_\parallel$ obeying

$$\frac{Q_2}{D} = \frac{2y}{5[3\lambda(y) - y]}, \quad \lambda(y) = 1 + (y-1)e^{-y}, \quad s_e = y(1 - e^{-y}), \quad (2)$$

for any isotropic filter and any source amplitude (the division-free form and the anisotropic control that violates it are in the lead's report). At $y = 1$ the ratio is $1/5$; as $y \rightarrow 0$ with external dominance kept it tends to $2/25$. Independent check: the nonlinear axisymmetric machinery of the Solar-System gate (log-radius by Legendre grid, mode-wise output filter) gives $Q_2/D = 1.0931$ against 1.1089 at $y = 2.478$ (canonical) and 0.6137 against 0.6102 at $y = 2.057$ (alternate), constant to 5×10^{-5} across $\xi = 0.3\text{--}10$ pc and source masses 1 and $10^{-3} M_\odot$. This cross-check also exposed and fixed a wrong-direction external-field conversion in the first Solar-System scan.

3 Both sources varied

With $v_\xi(x) = -\text{erf}(|x|/\sqrt{2}\xi)/|x|$ and its Hessian H_ξ , independent variation of the source positions gives the phantom acceleration of body a ,

$$\mathbf{a}_{a,\text{ph}} = -\frac{1}{4\pi} \int H_\xi(x - x_a) [f(p_e + p) - f(p_e)] d^3x, \quad f(p) = [\nu(|p|/a_0) - 1] p, \quad (3)$$

equivalent to $F_i = -\int \rho_{\text{ph}} g_i^{(S)}$ with the output filter moved onto the body by self-adjointness. Results, all from committed scripts with checks that can fail:

- Isolated compact filtered pair ($d \ll \xi$, $\xi \gg r_M$): the relative equation is $\ddot{\mathbf{r}} = -GM\mathbf{r}/r^3 - h(M)\mathbf{r}$ with $h = \frac{2}{9}\sqrt{GMa_0}/\xi^2$ at leading order; direct quadrature gives 0.95–0.96 of this at $\xi/r_M \approx 8$ and 0.98 at 25, independent of d/ξ between 0.1 and 0.2 to 0.5%.
- Isolated pair at $\xi = 0$: Milgrom's exact deep-MOND two-body force $Fd = \frac{2}{3}\sqrt{Ga_0}[M^{3/2} - M_1^{3/2} - M_2^{3/2}]$ reproduced to 1.0% (equal masses) and 0.8% (10:1) at $d = 100 r_M$; $-\partial E/\partial d$ equals the direct force to 0.05–0.3%; third law to 0.1% of the total force.
- Theorem (QUMOND stress tensor $T_{ij} = \partial_j u \partial_i \Phi + \partial_j \Phi \partial_i u - \nu \partial_j u \partial_i u - \delta_{ij}(\nabla u \cdot \nabla \Phi - \frac{1}{2}a_0^2 Q)$): the total force on bounded matter in a uniform external Newtonian field is $\nu_e g_e M_{\text{tot}}$ exactly, so the phantom exerts no net force on the bodies. Verified to 2×10^{-4} ($\xi = 0$) and 10^{-5} ($\xi = 0.03$ pc); holds for the filtered action by translation invariance.

- External-field regime: the identity becomes $2y(\omega_r^2 - \omega_\phi^2) - 3[5\lambda(y) - y](\omega_\phi^2 - \omega_z^2) = 0$ for infinitesimal radial and vertical perturbations of a circular orbit perpendicular to the field axis (harmonic-core, linear-response limit; equal to 6 at $y = 1$). It is an orbital translation of the tidal identity, not new dynamics.
- Aligned wide binaries in the Galactic field ($g_{\text{obs}} = 2.32 \times 10^{-10}$, converted to the Newtonian gradient by the spherical relation): the total-mass one-body reduction errs by $< 2.5\%$ at 4–30 kAU except an 8–10% spike at $d = \sqrt{GM_{\text{tot}}/g_e}$ (7 kAU for $2M_\odot$, 5 kAU for $1M_\odot$), the on-axis field null. At the Solar-System floors the double filter holds the aligned boost below 1% inside 15 kAU. The perpendicular orientation is not computed; these are not γ_v predictions.

4 Solar-System gates

Signed quadrupole against the Park et al. (2026) two-sigma ceiling $5.2 \times 10^{-27} \text{ s}^{-2}$, phantom mass inside Saturn’s orbit against Pitjev–Pitjeva, and the radial anomaly at Mercury–Neptune against the repository’s $\alpha = 1$ sunward gate, at three external-field inputs: the double filter survives for $\xi \geq 0.02$ pc (Gaussian, canonical), 0.03 pc (Gaussian alternate; Helmholtz both). The strict exponential AQUAL law without a length fails the quadrupole by $4\text{--}5\times$ on both footings by two independent discretisations. A single-response Helmholtz proxy for the local operator of §6 survives for $\xi \geq 0.03/0.05$ pc. These are tabulated grid points under inherited observational inputs, not continuum boundaries.

5 Toward a covariant action

C-H (the lead’s G03). An explicit action with metric, a dynamical clock τ (unit normal $n_\mu = -\partial_\mu \tau / \sqrt{-(\partial\tau)^2}$), an auxiliary potential U , a heat coordinate z realising the filter as $\partial_z W = \Delta_h W$ on the leaves, two multipliers, Λ , minimally coupled matter and temporal boundary terms; gravitational modification $2|DU - a|^2 + 2\alpha^2 q(|DW_b|^2/\alpha^2) + \int_0^{\xi^2/2} dz L(\partial_z W - \Delta_h W) + \lambda_0(W_0 - U)$, $\alpha = a_0/c^2$. Established: eliminating the spatial metric potential at leading weak static order gives exactly the filtered functional with both filters; $W_0 = U$, $W_b = SU$ and the weighted adjoint $N^{-1}SN$ are derived; 13/13 finite and symbolic checks (nonlinear first variations to 6×10^{-11} , with the outer filter’s removal producing 0.03–0.08 errors). Open: the physical mode count of the clock/metric system, causal feasibility of the heat sector (the naive scalar-wave realisation fails), $\Phi = \Psi$ beyond leading order, the isolated $\mathbb{R}^3$ limit.

An intrinsic-elliptic Deffayet–Woodard variant is dead: its action forces $R_{\mu\nu}\hat{u}^\mu\hat{u}^\nu = 0$ on every homogeneous solution, hence $\ddot{a} = 0$: no de Sitter, radiation, matter or Λ CDM background.

6 The operator that screens the Solar System

In an aether-scalar host the preferred-frame parameter α_1 carries an untunable scalar drag; only a *coherent* stiffening of the scalar’s whole quadratic form, $J_Y \rightarrow J_Y(1 + \xi^2 k^2)$, suppresses it, and local fourth-order operators on the scalar alone fail. The term

$$Y \rightarrow Y + \xi^2 q^{\lambda\sigma} q^{\mu\nu} \nabla_\lambda V_\mu \nabla_\sigma V_\nu, \quad V_\mu = q_\mu^\nu \partial_\nu \phi, \quad q_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu, \quad (4)$$

does it from an action: V ’s background vanishes, so at quadratic order $(\nabla_\perp V)^2 = [k^2 + (n \cdot k)^2] V_1 \cdot V_1$ multiplies the whole Y sector, mixings included. In the repository’s boosted PPN pipeline (Will

dictionary, static ladder to second order in the boost) α_1 of the operator equals the coherent-stiffening value $-4(2 - K_B)/(J_Y(1 + \xi^2 k^2) + 1)$ plus the host's own $-4c_{14}$ exactly at every ladder point, in the aether host and in the hypersurface-orthogonal clock host (aether replaced by $-\partial\tau/|\partial\tau|$ to first order, transverse components included). With $c_2 = c_4 = 1/10$ the scalar's α_2 drag is suppressed from 16.5 to 5.9×10^{-4} (aether) and 5.7 to 1.8×10^{-4} (clock) at $(\xi k)^2 = 10^4$; in the khronometric corner $c_{13} = 0$, $c_{14} = 10^{-5}$, $c_2 = 0.1$ the clock's own $\alpha_1 = -4.7 \times 10^{-5}$ and $\alpha_2 = -5.9 \times 10^{-6}$, and $\gamma = 1$, $\alpha_3 = 0$ throughout. At the Cassini floor $(\xi k)^2 \geq 8.6 \times 10^7$ at 1 AU, every scalar drag is below 10^{-7} . The operator is purely spatial in the clock frame: one degree of freedom, positive Hamiltonian, dispersion $\omega^2 = c_s^2 k^2(1 + \xi^2 k^2)$ (the healing-length form), group velocity $2\xi k c_s$ at short wavelength. The last number is what the causality gate must judge; placed outside the MOND function with its own coefficient the same term makes the static operator uniformly elliptic at zero field, which is a separate, uncomputed claim.

7 What is not claimed

No completed relativistic theory; ξ is a constraint (0.02–0.05 pc from the Solar System, 15–20 kAU knee for wide binaries, 50–140 pc wanted by three outer-halo globular clusters) not a derivation; $a_0 = \frac{1}{2}c\sqrt{G\rho_\Lambda}$ is an optional relation; the exponential and RAR kernels remain undecided on galaxy data once a_0 and mass-to-light are profiled; the clock corner's health with the scalar present, FLRW, the measured G and the zero-field limit are open; the wide-binary registration and its hashes are unchanged.

Reproduction

Repository <https://github.com/carlzimmerman/zimmerman-formula>, `qwen_claude_field_theory/closure_2026`, `filtered_tidal_relation_2026`, `two_body_frequency_2026`, `g03_covariant_action_2026`, `dw_intrinsic_ell` (lead), `g01_strict_aqual.py`, `g02_filtered_efe.py`, `g02b`, `g02c_two_body_force.py`, `g02d`, `g03b`, `g03pre`; `hunt_2026/f31c`, `f32`, `f33`. Each script exits nonzero on a failed check. Observational inputs: Park et al. 2026 (Cassini Q_2), Pitjev & Pitjeva 2013, the repository's ephemeris gate.

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